

Exercise set 1

- 1) Why do we need computation? Why do we need programming language for computation? Why do we need automats / machines that recognize/accept programming language?

- 2) For any $n \in \mathbb{N}$, prove that the following equality is valid.

$$1^6 + 2^6 + 3^6 + \dots + n^6 = \frac{n}{42}(n+1)(2n+1)(3n^4 + 6n^3 - 3n + 1)$$

3) Let α and β be two positive integer numbers. If $\alpha^2 - \beta^2$ is not odd, then prove that $\alpha + \beta \geq 2$.

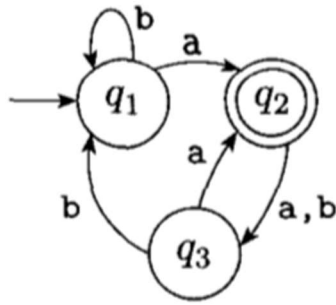
4) Let a, b, c, d be integers. If $a > c$ and $b > c$, then prove that $\max(a, b) - c$ is always positive.

5) Given two sets X and Y . The Cartesian product of X and Y , written as $X \times Y$, is defined as the set of pairs (x, y) where $x \in X$ and $y \in Y$. Then, find a mathematical closed-form expression to write $|X \times Y|$ in terms of $|X|$ and $|Y|$.

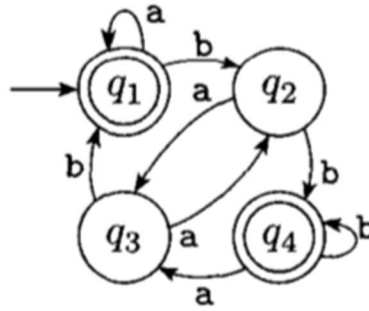
6) Let us given two disjoint sets X and Y , and then their joint set S is $S = X \cup Y$. Sum of the elements in a set S is denoted by $\sum(S)$ while their product is by $\prod(S)$. Accordingly, what are $\sum(S)$ and $\prod(S)$ in terms of $\sum(X)$, $\sum(Y)$, $\prod(X)$ and $\prod(Y)$. Conclude from this what $\sum(\emptyset)$ and $\prod(\emptyset)$ should be ($\emptyset$ is the empty set).

7) What is the relation between programming language and the power of a machine that recognizes / accepts that programming language? Give an example in your explanation.

- 8) The following are the state diagrams of two DFAs, M_1 and M_2 . Answer the following questions about each of these machines.



M_1



M_2

- What is the start state?
- What is the set of accept states?
- What sequence of states does the machine go through on input abb ?
- Does the machine accept the string abb ?
- Does the machine accept the string ϵ ?

9) The formal 5-tuple description of a DFA M is

$$(\{q_1, q_2, q_3, q_4, q_5\}, \{u, d\}, \delta, q_1, \{q_3\}),$$

where δ is given by the following table. Give the state diagram of this machine.

	u	d
q_1	q_1	q_2
q_2	q_1	q_3
q_3	q_2	q_4
q_4	q_3	q_5
q_5	q_4	q_5

10) Give state diagrams of DFAs recognizing the following languages. In all parts the alphabet is $\{0,1\}$.

a) $\{w \mid w \text{ begins with a 1 and ends with a 0}\}$

b) $\{w \mid w \text{ contains at least three 1s}\}$

c) $\{w \mid w \text{ contains the substring 0101, i.e., } w = x0101y \text{ for some } x \text{ and } y\}$

d) $\{w \mid w \text{ has length at least 3 and its third symbol is a 0}\}$

e) $\{w \mid w \text{ starts with 0 and has odd length, or starts with 1 and has even length}\}$

f) $\{w \mid w \text{ doesn't contain the substring 1101}\}$

g) $\{w \mid \text{the length of } w \text{ is at most } 5\}$

h) $\{w \mid w \text{ is any string except } 11 \text{ and } 1111\}$

i) $\{w \mid \text{every odd position of } w \text{ is a } 1\}$

j) $\{w \mid w \text{ contains at least two 0s and at most one 1}\}$

k) $\{\epsilon, 0\}$

l) $\{w \mid w \text{ contains an even number of 0s, or contains exactly two 1s}\}$

m) The empty set

n) All strings except the empty string

11) Show by giving an example that if M is a DFA that recognizes language C , swapping the final and non-final states in M yields a new DFA that recognizes $\bar{C}$.

12) Design automata (DFA) to accept the following languages:

a) $A = \{w \in \{0, 1\}^* : w \text{ has a } 1 \text{ in the third position from the right}\}.$

b) $B = \{w \in \{0, 1\}^* : w \text{ contains at least two } 0\text{s}\}$

c) $C = \{w \in \{0, 1\}^* : \text{the length of } w \text{ is divisible by three}\}$

d) $D = \{w \in \{0, 1\}^* : w \text{ contains exactly two 0s and at least two 1s}\}.$

13) Give state diagrams of NFAs with the specified number of states recognizing each of the following languages. In all parts the alphabet is $\{0,1\}$.

a) The language $\{w \mid w \text{ ends with } 00\}$ with three states

b) The language $\{w \mid w \text{ contains the substring } 0101, \text{ i.e., } w = x0101y \text{ for some } x \text{ and } y\}$ with five states

c) The language $\{w \mid w \text{ contains an even number of 0s, or contains exactly two 1s}\}$ with six states

d) The language $\{0\}$ with two states

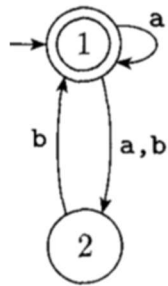
e) The language $0^*1^*0^+$ with three states

f) The language $1^*(001^+)^*$ with three states

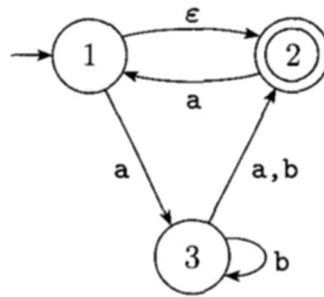
g) The language $\{\varepsilon\}$ with one state

h) The language 0^* with one state

- 14) Use the construction given in Theorem 1.39 in the book to convert the following two non-deterministic finite automata to equivalent deterministic finite automata.

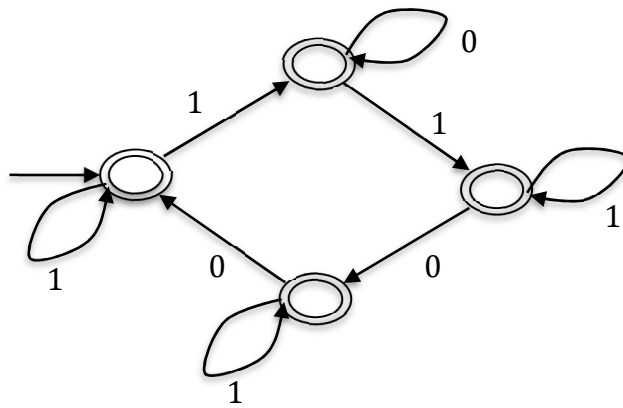


(a)



(b)

- 15) For the alphabet $\Sigma_1 = \{0,1\}$, answer the following questions for the automata machine shown below



a) Is the machine DFA or NFA? Why?

b) Give its regular expression.

c) Write the language which is a set consisting of strings that are recognized by this automaton.

16) Give regular expressions describing the following languages:

a) $A = \{w \in \{0,1\}^* : w \text{ contains at least three } 1s\}$.

b) $B = \{w \in \{0,1\}^* : w \text{ contains at least two } 1s \text{ and at most one } 0\}$,

c) $C = \{w \in \{0,1\}^* : w \text{ contains an even number of } 0s \text{ and exactly two } 1s\}$.

d) $D = \{w \in \{0, 1\}^* : w \text{ contains an even number of 0s and each 0 is followed by at least one } 1\}$

17) Design a DFA or NFA for the following languages. $n_0(w)$ denotes the number of zeros in the string w .

a) $L_1 = \{w \in \{0, 1\}^* : n_0(w) \bmod 2 = 0\}$,

b) $L_2 = \{w \in \{0, 1\}^* : n_0(w) \bmod 3 = 0\}$,

c) Based on using the NFA and DFA you designed in the options a and b, design an NFA that recognized the language $L_3 = \{w \in \{0, 1\}^* : n_0(w) \bmod 6 = 0\}$.

Hint: De Morgan's Laws $L_1 \cap L_2 = \overline{(\overline{L_1} \cup \overline{L_2})}$ can be used for designing an NFA that recognizes the intersection of languages.